

Multiple Choice (10 marks)

1. Which of the following does authorization aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
2. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
3. _____ is a widespread app's coding mistake made by developers which could be exploited by an attacker for gaining access or malfunctioning your system.
 - (a) Memory leakage
 - (b) Buffer-overflow
 - (c) Less processing power
 - (d) Inefficient programming
4. Which of the following is true of mutation-based fuzzing?
 - (a) It generates each different input by modifying a prior input
 - (b) It works by making small mutations to the target program to induce faults
 - (c) Each input is mutation that follows a given grammar
 - (d) It only makes sense for file-based fuzzing, not network-based fuzzing
5. Three of the following are classic security properties; which one is not?
 - (a) Confidentiality
 - (b) Availability
 - (c) Correctness
 - (d) Integrity

True / False (10 marks)

Answer True or False

6. True or False: A packet filter firewall operates primarily at the network or transport layer of the OSI model.
7. True or False: A stack is a sequential segment of memory used for storing data like character strings or arrays of integers.
8. True or False: Blackbox fuzzers can analyze the internal structure of a program to ensure complete code coverage like whitebox fuzzers do.
9. True or False: The Morris Worm is widely recognized as the first buffer overflow attack in cybersecurity history.
10. True or False: Backdoor Trojans are primarily used to encrypt files and demand ransom, rather than stealing instant messaging credentials.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check for majority element in the given sorted array.
12. Write a python function to check whether the given two arrays are equal or not.

End of Paper